

Literature Review

Chapter 2 — Literature Review

第 2 章 — 文献综述

This chapter reviews the literature needed to position this dissertation. Section 2.1 reviews crude-oil price forecasting and explains why the random-walk benchmark and financial baselines are difficult to beat. Sections 2.2 and 2.3 examine two alternative-data sources — maritime/AIS shipping activity and satellite remote sensing — as economic proxies for oil-market conditions. Section 2.4 reviews multimodal forecasting and the distinction between flat feature fusion and representation-level modality-aware fusion. Section 2.5 reviews forecast evaluation and interpretability standards. Section 2.6 synthesises these strands into the research gap addressed by this dissertation.

本章综述与本论文研究定位直接相关的文献。第 2.1 节回顾原油价格预测研究，并解释为什么随机游走基准和金融基准模型很难被超越。第 2.2 节和第 2.3 节分别讨论两类替代数据来源——海运/AIS 航运活动和卫星遥感——如何作为原油市场状况的经济代理变量。第 2.4 节回顾多模态预测研究，并区分扁平特征融合与表示层面的模态感知融合。第 2.5 节讨论预测评估与可解释性标准。第 2.6 节综合上述文献，提出本文所关注的研究空白。

2.1 Crude-oil price forecasting

2.1 原油价格预测

2.1.1 Econometric foundations and the benchmark problem

2.1.1 计量经济学基础与基准问题

Oil prices are difficult to predict out of sample. Kilian (2009) [P052] shows that price movements should be understood through different structural channels, including crude-oil supply shocks, aggregate-demand shocks and oil-specific precautionary-demand shocks. The decomposition is not itself a forecasting model, but it does offer a useful principle for predictor selection: useful variables should have a plausible economic connection to supply, demand or uncertainty, rather than being included only because they are available.

原油价格在样本外很难被准确预测。Kilian (2009) [P052] 指出，原油价格变动应从不同结构性渠道理解，包括原油供给冲击、总需求冲击和石油特定的预防性需求冲击。该分解本身并不是一个预测模型，但它为预测变量选择提供了一个有用原则：有用的变量应当与供给、需求或不确定性具有合理的经济联系，而不是仅仅因为数据可得就被纳入模型。

Alquist, Kilian and Vigfusson (2013) [P053] provide the key forecasting benchmark for this dissertation. They show that the no-change, or random-walk, forecast is extremely difficult to beat in oil-price forecasting, especially out of sample. Their review also emphasises that in-sample fit does not imply forecasting skill, and that claims of predictability should be tested using real-time data alignment, recursive or rolling evaluation and formal forecast-comparison tests. Baumeister and Kilian (2015) [P054] further show that forecast combinations across different economic mechanisms can be more robust than reliance on a single predictor set. Taken together, these studies set a high bar: any alternative-data model must be evaluated against a strong no-change benchmark and an economically informed financial baseline, not only against other machine-learning models.

Alquist, Kilian and Vigfusson (2013) [P053] 为本文提供了关键的预测基准。他们表明，无变化预测，即随机游走预测，在原油价格预测中极难被超越，尤其是在样本外预测中。他们的综述还强调，样本内拟合并不等同于预测能力；任何关于可预测性的主张都应基于实时数据对齐、递归或滚动窗口评估，以及正式的预测比较检验。Baumeister and Kilian (2015) [P054] 进一步指出，跨不同经济机制的预测组合往往比依赖单一预测变量集更加稳健。综合来看，这些研究设定了很高的检验门槛：任何替代数据模型不仅需要与其他机器学习模型比较，还必须与强随机游走基准和具有经济含义的金融基准模型比较。

2.1.2 Machine-learning approaches to oil-price forecasting

2.1.2 原油价格预测中的机器学习方法

Machine-learning studies have introduced tree ensembles, regularised linear models, deep learning and hybrid models into oil-price forecasting. Costa et al. (2021) [P072] compare a broad set of methods over a large macro-financial predictor set and find that useful predictors vary across horizons and over time. XGBoost can be a strong benchmark, but it does not dominate uniformly. Yilmaz and Zehir (2026) [P076] show that geopolitical risk, market volatility and interest-rate variables can add value for Brent-return forecasting, with LightGBM outperforming XGBoost in their setting. Foroutan and Lahmiri (2024) [P001] report strong performance from temporal convolutional networks and gradient-boosting models, but their focus on price-level prediction also illustrates a common problem: because oil prices are highly persistent, low one-step price-level errors can partly reflect the fact that P_{t+1} is usually close to P_t .

机器学习研究已经将树模型、正则化线性模型、深度学习和混合模型引入原油价格预测。Costa et al. (2021) [P072] 在大规模宏观金融预测变量集上比较了多种方法，发现有用预测变量会随预测期限和时间变化而变化。XGBoost 可以构成强有力的基准，但并不在所有情境下都占优。Yilmaz and Zehir (2026) [P076] 表明，地缘政治风险、市场波动率和利率变量能够为 Brent 收益率预测提供增量信息，其中 LightGBM 在他们的设定中优于 XGBoost。Foroutan and Lahmiri (2024) [P001] 报告称，时间卷积网络和梯度提升模型表现较强，但其对价格水平预测的关注也反映出一个常见问题：由于原油价格高度持久，较低的一步价格水平误差可能部分来自 P_{t+1} 通常接近 P_t 这一事实。

Several recent studies use more complex architectures, but their results should be read cautiously. Hybrid designs such as LSTM feature extraction combined with XGBoost have reported very high R^2 values [P004], yet such results may be sensitive to preprocessing choices such as pre-split normalisation. Graph-based oil-price models have also appeared. Zhao, Xue and Cheng (2023) [P063], for example, combine a self-attention-learned dynamic graph with Graph WaveNet (Wu et al., 2019) [P091] for multi-step WTI futures forecasting. Their graph represents non-Euclidean relations among predictors rather than a physical shipping or geographic network, and the absence of a no-change benchmark makes it difficult to assess whether the model improves over a random walk. I treat these papers as evidence that model complexity alone is not enough; they are background rather than direct templates for the weekly Brent forecasting problem addressed here.

一些近期研究采用了更复杂的模型结构，但其结果需要谨慎解读。例如，将 LSTM 特征提取与 XGBoost 相结合的混合设计报告了极高的 R^2 [P004]，但这类结果可能对预处理选择非常敏感，例如是否存在数据划分前的标准化泄漏。图模型也开始出现在原油价格预测中。Zhao, Xue and Cheng (2023) [P063] 将自注意力学习到的动态图与 Graph WaveNet (Wu et al., 2019) [P091] 结合，用于多步 WTI 期货预测。其图结构表示的是预测变量之间的非欧几里得关系，而不是物理意义上的航运网络或地理网络；同时，由于缺少无变化基准，模型是否真正优于随机游走仍难以判断。我将这些研究视为说明“模型复杂度本身并不足够”的证据；它们是背景文献，而不是本文周度 Brent 预测问题的直接设计模板。

For this dissertation, three points from this literature matter. The price's own lag and the random-walk forecast must be treated as serious competitors, not weak baselines. Financial and oil-market fundamentals provide an economically meaningful baseline before alternative data are added: alongside volatility, geopolitical risk, interest rates, exchange rates and market-based oil indicators, inventories, production and refinery activity are natural proxies for Kilian's supply and demand channels. Any claim that shipping or remote sensing improves forecasting must rest on incremental out-of-sample value over that baseline, not on raw error reduction alone.

对本文而言，该文献中有三点尤其重要。第一，价格自身滞后项和随机游走基准必须被视为强竞争基准，而不是弱基准。第二，金融与原油市场基本面应在加入替代数据前构成具有经济意义的金融基准：除波动率、地缘政治风险、利率、汇率和市场型原油指标外，库存、产量、炼厂活动等实物供需变量也是 Kilian 供给与需求渠道的天然代理。第三，任何关于航运或遥感数据提升预测能力的主张，都应建立在相对于该金融基准的样本外增量价值之上，而不能仅依赖原始误差下降。

2.2 Shipping activity and oil markets

2.2 航运活动与原油市场

2.2.1 AIS and maritime activity as trade-flow proxies

2.2.1 AIS 与海运活动作为贸易流代理变量

Automatic Identification System (AIS) vessel tracking has become an important high-frequency proxy for physical trade. Adland et al. (2017) [P014] validate AIS-derived crude-export estimates against official statistics, while Yan et al. (2020) [P015] show that global marine oil trade is concentrated around major chokepoints such as Hormuz, Malacca and Suez. Arslanalp, Marini and Tumbarello (2019) [P018] and the IMF PortWatch methodology [P070] further demonstrate how vessel movements can be used to nowcast trade activity. This literature supports treating shipping data as a plausible source of physical-market information, without assuming that AIS counts directly measure crude flows or predict prices.

自动识别系统（Automatic Identification System, AIS）船舶轨迹数据已经成为衡量实物贸易活动的重要高频代理变量。Adland et al. (2017) [P014] 将基于 AIS 估算的原油出口量与官方统计数据进行了验证；Yan et al. (2020) [P015] 则表明，全球海上原油贸易高度集中于霍尔木兹海峡、马六甲海峡和苏伊士运河等关键咽喉。Arslanalp, Marini and Tumbarello (2019) [P018] 以及 IMF PortWatch 方法 [P070] 进一步展示了如何利用船舶移动数据对贸易活动进行即时预测。这些文献支持将航运数据视为实物市场信息的合理来源，但并不意味着 AIS 计数能够直接衡量原油流量或预测价格。

Capacity-weighted indicators and draught-change measures are generally more informative than simple vessel counts, because they better approximate cargo movement. AIS data also require careful filtering to exclude non-trade activity, and moving averages must be constructed without using future observations. A shipping indicator is useful for forecasting only if it is both economically meaningful and available at the forecast origin. In practice, that means chokepoint transit volumes and tanker-presence measures are better read as activity proxies at key maritime nodes than as direct measures of global supply.

与简单船舶数量相比，按运力加权的指标和吃水变化指标通常更具信息含量，因为它们更接近货物流动本身。AIS 数据也需要仔细过滤，以排除非贸易活动；移动平均等处理也必须避免使用未来观测值。一个航运指标只有在其既具有经济含义、又在预测时点真实可得时，才可能对预测有用。在实践中，这意味着咽喉通行量和油轮存在度指标更适合被理解为关键海运节点的活动代理，而不是全球供给的直接测量。

2.2.2 Reverse causality and proxy limitations

2.2.2 反向因果与代理变量限制

The relationship between shipping activity and oil prices is not one-directional. Mi et al. (2022) [P016] and Mi, Zang, Lo and Chen (2023) [P017] study how crude-oil prices affect tanker port-call activity, rather than how tanker activity predicts prices. Their findings indicate that the relationship is non-linear and regionally heterogeneous, and that statistically significant relationships may explain only a small share of variation. Shipping data may contain useful information about physical trade and congestion, but they may also respond to oil prices rather than lead them.

航运活动与原油价格之间的关系并不是单向的。Mi et al. (2022) [P016] 以及 Mi, Zang, Lo and Chen (2023) [P017] 研究的是原油价格如何影响油轮港口停靠活动，而不是油轮活动如何预测价格。他们的发现表明，这一关系具有非线性和区域异质性，统计显著的关系可能只能解释较小比例的变异。航运数据可能包含关于实物贸易和拥堵的有用信息，但它也可能是对原油价格变化的反应，而非领先信号。

Chokepoint and port indicators are also imperfect proxies for crude-oil flows. A vessel's previous port is not always the cargo origin; crude oil may be blended or re-sold; and ship-to-ship transfers can obscure the true trade route. Paolo et al. (2024) [P057] further show that a substantial amount of industrial activity at sea is absent from AIS. Shipping indicators should therefore be interpreted as noisy proxies for physical-market conditions, not direct measurements of global oil supply.

咽喉和港口指标也只是原油流动的不完美代理。船舶的上一港口并不总是货物原产地，原油可能被混合、转售，船对船转运也可能模糊真实贸易路径。Paolo et al. (2024) [P057] 进一步表明，大量海上工业活动并未被 AIS 覆盖。因此，航运指标应被理解为对实物市场状况的噪声代理变量，而不是全球原油供给的直接测量。

2.2.3 From flat shipping indicators to maritime structure

2.2.3 从扁平航运指标到海运结构

Most oil-related uses of AIS and PortWatch-style data convert shipping activity into tabular features such as port calls, vessel counts or chokepoint transit volumes. This is useful, but it discards the network structure of maritime activity. Maritime trade is inherently spatial and relational: ports, terminals, chokepoints and routes form a connected transport system. Studies such as Ouyang et al. (2022) [P062] and Liang et al. (2022) [P066] show that graph-based models can learn spatial-temporal structure in vessel-flow prediction. Those studies forecast traffic flows rather than oil prices. They justify the idea that maritime structure is learnable, but they do not prove that a maritime representation improves Brent-price forecasting.

多数与原油相关的 AIS 和 PortWatch 数据应用，会将航运活动转化为港口停靠数、船舶数量或咽喉通行量等表格型特征。这种做法有实际价值，但会丢失海运活动本身的网络结构。海上贸易本质上具有空间性和关系性：港口、码头、咽喉和航线共同构成一个相互连接的运输系统。

Ouyang et al. (2022) [P062] 和 Liang et al. (2022) [P066] 等研究表明，图模型可以学习船舶流量预测中的时空结构。这些研究预测的是交通流，而不是原油价格。它们支持“海运结构可以被学习”这一判断，但并没有证明海运表示一定能提升 Brent 价格预测。

2.3 Satellite imagery and remote sensing

2.3 卫星影像与遥感

2.3.1 Remote sensing as an economic proxy

2.3.1 遥感作为经济代理变量

Remote sensing provides a physical view of economic activity, infrastructure and environmental conditions. In oil-related applications, night-time lights, NO₂, cloud cover and high-resolution imagery have all been used as indirect indicators of economic activity, trade, demand or inventory information. The literature is cautious about what these signals can and cannot measure.

遥感数据提供了观察经济活动、基础设施和环境条件的物理视角。在原油相关应用中，夜间灯光、NO₂、云量和高分辨率影像都曾被用作经济活动、贸易、需求或库存信息的间接指标。相关文献对这些信号能够衡量什么、不能衡量什么持较为谨慎的态度。

Night-time lights are one of the most widely used remote-sensing proxies, but their usefulness depends on scale. Polinov, Bookman and Levin (2022) [P024] find that night-time lights correlate with anchorage activity at a broad cross-sectional scale, yet they do not reliably track tanker activity at a single port. Gibson et al. (2021) [P032] similarly show that VIIRS night-time lights are more suitable than DMSP for facility-scale work, but that night lights capture cross-sectional differences more reliably than within-unit temporal variation. Raw radiance should not be treated as a direct time-series measure of oil activity. Within-site anomalies are more defensible than raw levels when the research objective is forecasting over time.

夜间灯光是最常用的遥感代理变量之一，但其有效性依赖于空间尺度。Polinov, Bookman and Levin (2022) [P024] 发现，夜间灯光在较大的横截面尺度上与锚地活动相关，但并不能可靠追踪单一港口的油轮活动。Gibson et al. (2021) [P032] 同样表明，VIIRS 夜间灯光比 DMSP 更适合设施尺度研究，但夜间灯光更擅长捕捉横截面差异，而非单位内部的时间变化。原始辐亮度不应被简单视为原油活动的直接时间序列指标。当研究目标是时间序列预测时，站点内部异常值比原始水平值更具合理性。

Other remote-sensing indicators provide different mechanisms. Hao and Wang (2023) [P025] link cloud cover over US storage regions to next-week WTI returns through an information-availability channel: when clouds obstruct optical observation of storage tanks, market uncertainty about inventories may increase. Bricongne et al. (2026) [P069] use tropospheric NO₂ to nowcast national oil demand, but their results also show that the incremental value of remote-sensing variables can weaken inside non-linear

models. Wang et al. (2019) [P055] estimate oil-tank structural capacity from high-resolution images, which supports infrastructure measurement but not high-frequency inventory estimation at Sentinel-2 resolution.

其他遥感指标对应不同机制。Hao and Wang (2023) [P025] 将美国储油区上空云量与下一周 WTI 收益率联系起来，其机制是信息可得性：当云层遮挡储油罐的光学观测时，市场关于库存的不确定性可能上升。Bricongne et al. (2026) [P069] 使用对流层 NO₂ 对国家层面的石油需求进行即时预测，但他们的结果也显示，当进入非线性模型后，遥感变量的增量价值可能减弱。Wang et al. (2019) [P055] 利用高分辨率影像估计储油罐结构容量，这支持基础设施测量，但并不等同于在 Sentinel-2 分辨率下进行高频库存估计。

2.3.2 Limits of direct RS-to-price claims

2.3.2 直接 RS-to-price 主张的限制

Satellite data may contain oil-relevant information, but the literature does not support a simple claim that satellite imagery directly predicts oil prices. Most remote-sensing indicators are upstream proxies: they may reflect industrial activity, port activity, storage observability or demand conditions, which may then influence prices through supply-demand expectations. The mechanism is indirect and may vary across locations, sensors and time horizons.

卫星数据可能包含与原油市场相关的信息，但文献并不支持“卫星影像可以直接预测原油价格”的简单主张。多数遥感指标是上游代理变量：它们可能反映工业活动、港口活动、库存可观测性或需求条件，而这些因素又可能通过供需预期影响价格。该机制是间接的，并且可能随地点、传感器和预测期限而变化。

Jung (2026) [P068] provides a useful example of this limitation. The study combines satellite-derived indicators with port attributes to nowcast port-level trade, but the target is trade rather than price, and the model relies on engineered tabular features rather than learned image representations. This is representative of a broader pattern: remote sensing is often converted into flat numeric columns before entering an economic model. Such features may be valuable, but what remains unclear is whether preserving image or site-level representations adds forecasting value for weekly Brent prices.

Jung (2026) [P068] 提供了一个有用例子。该研究将卫星衍生指标与港口属性结合，用于即时预测港口层面的贸易，但其预测目标是贸易而非价格，并且模型依赖工程化表格特征，而不是学习到的图像表示。这代表了更广泛的模式：遥感数据通常先被压缩为扁平数值列，再进入经济模型。这些特征可能具有价值，但尚不清楚的是，保留图像或站点层面的表示是否能够提升周度 Brent 价格预测。

2.4 Multimodal forecasting and fusion

2.4 多模态预测与融合

2.4.1 From multi-source data to multimodal learning

2.4.1 从多源数据到多模态学习

A key distinction in this dissertation is between multi-source feature fusion and multimodal representation learning. Baltrušaitis, Ahuja and Morency (2019) [P101] define multimodal learning as involving representation, translation, alignment, fusion and co-learning. Within this taxonomy, fusion can occur at the feature level, decision level or representation level. In much of the oil-price forecasting literature, heterogeneous data are combined through early feature-level fusion: financial indicators, shipping counts and satellite-derived indices are concatenated into a single table and passed to a conventional model. This approach is practical, but it treats all inputs as ordinary numeric predictors and may discard modality-specific structure.

本文的一个关键区分是：多源特征融合并不等同于多模态表示学习。Baltrušaitis, Ahuja and Morency (2019) [P101] 将多模态学习定义为涵盖表示、转换、对齐、融合和协同学习等问题。在该分类框架下，融合可以发生在特征层、决策层或表示层。在多数原油价格预测研究中，异质数据通常通过早期特征层融合被合并：金融指标、航运计数和卫星衍生指数被拼接成一个表格，然后输入常规模型。这种方法具有实践便利性，但它会将所有输入都视作普通数值预测变量，并可能丢失模态特有结构。

A modality-aware alternative is to encode different data types separately before fusing their representations. At the fusion step itself, learned gating offers a way to weight modalities dynamically rather than treating them equally: Arevalo et al. (2017) [P096] propose gated multimodal units in which the model learns, for each input, how much each modality contributes to the fused representation. Gohari et al. (2024) [P039] provide a relevant precedent in financial time-series forecasting, showing that modality-aware transformers can outperform naïve concatenation when combining different financial information sources. Their application involves text and numeric data rather than satellite imagery, maritime networks and oil-market variables. The study motivates modality-aware forecasting, but it does not answer whether such fusion is useful for crude-oil price prediction.

模态感知的替代方案是先分别编码不同数据类型，再融合它们的表示。在融合环节本身，可学习的门控提供了一种动态加权各模态、而非同等对待的机制：Arevalo et al. (2017) [P096] 提出门控多模态单元（gated multimodal units），让模型针对每个输入学习各模态对融合表示的贡献权重。Gohari et al. (2024) [P039] 在金融时间序列预测中提供了一个相关先例，表明在结合不同金融信息来源时，模态感知 Transformer 可以优于简单拼接。但该研究处理的是文本和数值数据，而不是卫星影像、海运网络和原油市场变量。它能够说明模态感知预测具有潜力，但并不能直接回答这种融合是否适用于原油价格预测。

2.4.2 Representation learning for Earth-observation data

2.4.2 地球观测数据的表示学习

Recent Earth-observation foundation models provide one route to transforming satellite imagery into representations. Models such as SatMAE [P095] and Prithvi-EO-2.0 [P094] use self-supervised pretraining to produce transferable image embeddings. Multisensor models such as CROMA [P105] further show that optical and radar data may benefit from modality-specific encoders before fusion, because the sensors differ in channel structure, noise and physical meaning. These models are relevant because weekly oil-price datasets are usually too small to train an image encoder from scratch.

近期地球观测基础模型为将卫星影像转化为表示提供了一条路径。SatMAE [P095] 和 Prithvi-EO-2.0 [P094] 等模型使用自监督预训练来生成可迁移的图像嵌入。CROMA [P105] 等多传感器模型进一步表明，光学和雷达数据在融合前可能需要模态特定编码器，因为不同传感器在通道结构、噪声和物理含义上存在差异。这些模型与本文相关，是因为周度原油价格样本通常太小，不足以从零开始训练图像编码器。

The EO foundation-model literature is mainly methodological rather than evidential for commodity-price forecasting. Most evaluations target land-cover classification, segmentation or related remote-sensing tasks, not economic forecasting or oil prices. The models show that satellite imagery can be represented by pretrained encoders; they do not show that such representations improve Brent forecasting.

地球观测基础模型文献对大宗商品价格预测而言，主要是方法论上的参考，而不是经验证据上的支撑。多数评估针对土地覆盖分类、语义分割或相关遥感任务，而不是经济预测或原油价格。这些模型说明卫星影像可以被预训练编码器表示，但并未证明这些表示能够改善 Brent 价格预测。

2.4.3 Missing and asynchronous modalities

2.4.3 缺失模态与异步模态

A further challenge is that alternative data sources are often incomplete or asynchronous. Optical satellite imagery is affected by cloud cover, radar and optical sensors have different revisit cycles, and shipping or macro-financial data may be released at different frequencies. General multimodal-learning studies show that models can degrade when modalities are missing unless missing-modality training or modality dropout is used [P097; P100], and related work on irregularly sampled time series further motivates the use of masks and time-since-observation signals [P098; P099].

另一个挑战是，替代数据来源通常存在缺失和时间不同步问题。光学卫星影像受云量影响，雷达和光学传感器具有不同重访周期，航运数据和宏观金融数据也可能以不同频率发布。一般多模态学习研究表明，如果不进行缺失模态训练或模态 dropout，模型在模态缺失时可能显著退化 [P097; P100]；针对不规则采样时间序列的相关研究也进一步说明，掩码和距上次观测时间等信号具有价值 [P098; P099]。

Missingness and temporal misalignment are part of the multimodal forecasting problem itself, not a secondary data-cleaning issue. Any fair comparison between flat feature fusion and representation-level fusion must account for the availability, timing and reliability of each modality.

缺失性和时间错位本身就是多模态预测问题的一部分，而不是次要的数据清洗问题。扁平特征融合与表示层融合之间的公平比较，必须考虑每个模态的可得性、时间对齐和可靠性。

2.5 Forecast evaluation and interpretability

2.5 预测评估与可解释性

2.5.1 Forecast comparison

2.5.1 预测比较

Because the random walk is difficult to beat, lower RMSE or MAE alone is not sufficient evidence of improved forecasting skill. Diebold and Mariano (1995) [P058] provide the standard framework for testing equal predictive accuracy across competing forecasts. For nested models, where one model extends another by adding predictors, Clark and West (2007) provide a more appropriate test under squared-error loss. These tests are especially relevant when evaluating alternative data, because the key question is not whether a large model can reduce error in one sample, but whether an added modality produces a statistically meaningful improvement over a simpler baseline.

由于随机游走很难被超越，仅仅报告较低的 RMSE 或 MAE 并不足以证明模型具有更强预测能力。Diebold and Mariano (1995) [P058] 提供了比较竞争预测之间预测精度是否相等的标准框架。对于嵌套模型，即一个模型通过增加预测变量扩展另一个模型的情形，Clark and West (2007) 在平方误差损失下提供了更合适的检验方法。这些检验在评估替代数据时尤其重要，因为关键问题不是一个大模型是否在某个样本中降低了误差，而是新增模态是否相对于更简单基准产生了统计上有意义的改进。

Alternative-data claims should therefore be evaluated through out-of-sample comparison, horizon-specific performance and formal tests of predictive accuracy. That standard applies whenever a financial baseline is compared with models that add shipping, remote sensing or multimodal representations.

替代数据的预测价值应通过样本外比较、分预测期限的表现和正式预测精度检验来评估。这一标准适用于将金融基准模型与加入航运、遥感或多模态表示的模型进行比较的任何情形。

2.5.2 Interpretability and modality-level explanation

2.5.2 可解释性与模态层面解释

Forecast accuracy tests can show whether a model improves, but they do not explain which signals drive the improvement. SHAP (Lundberg and Lee, 2017) [P059] provides one way to attribute model predictions to features and can be aggregated to the modality level. Modality-level explanation is

important because the dissertation is not only interested in whether the model forecasts better, but also whether financial, shipping and remote-sensing signals contribute in economically interpretable ways.

预测精度检验可以说明模型是否改进，但不能解释哪些信号推动了改进。SHAP (Lundberg and Lee, 2017) [P059] 提供了一种将模型预测归因于特征的方法，并且可以进一步聚合到模态层面。在本文情境中，模态层面解释很重要，因为论文不仅关心模型是否预测得更好，也关心金融、航运和遥感信号是否以经济上可解释的方式参与预测。

SHAP explains model behaviour; it does not establish causal effects. A high attribution for a shipping or satellite feature does not prove that the underlying physical activity caused the oil-price movement. Interpretability is best treated as complementary to forecast-comparison tests: accuracy tests assess whether a modality helps prediction, while attribution helps describe how the trained model uses that modality.

SHAP 解释的是模型行为，而不是因果效应。某个航运或卫星特征具有较高 attribution，并不意味着其背后的物理活动导致了原油价格变化。可解释性最好被视为预测比较检验的补充：精度检验评估某个模态是否有助于预测，而归因分析帮助描述训练后的模型如何使用该模态。

2.6 Synthesis, research gap and positioning

2.6 综合、研究空白与本文定位

2.6.1 Synthesis of the literature

2.6.1 文献综合

Four conclusions emerge from the review. Oil-price forecasting is difficult because oil prices are highly persistent and the random-walk benchmark is strong. Financial variables provide an essential economically informed baseline because they capture persistence, uncertainty, monetary conditions, exchange-rate channels and market expectations. Shipping and remote-sensing data are plausible alternative-data sources, but they are noisy and indirect proxies rather than direct measurements of future prices. Multimodal learning offers tools for preserving modality-specific structure, but these tools have not been systematically tested in the specific setting of weekly Brent forecasting.

综合上述文献，可以得出四点结论。第一，原油价格预测很困难，因为原油价格高度持久，随机游走基准很强。第二，金融变量构成必要的、有经济含义的基准，因为它们能够捕捉价格持续性、不确定性、货币条件、汇率渠道和市场预期。第三，航运和遥感数据是有潜力的替代数据来源，但它们是噪声较强、机制间接的代理变量，而不是未来价格的直接测量。第四，多模态学习为保留模态特定结构提供了工具，但这些工具尚未在周度 Brent 价格预测这一具体情境中得到系统检验。

The following table summarises the observable signal, economic channel and main limitation of each of the four literatures.

下表总结了四类文献各自的可观测信号、经济渠道与主要限制。

Data source / literature	Observable signal	Economic channel	Main limitation
Financial and oil-market variables	Lagged price, inventories, production/refinery activity, volatility, GPR, rates, exchange rates, futures/market indicators	Persistence, uncertainty, macro-financial conditions, market expectations	Strong benchmark; difficult to improve upon
Shipping / AIS / PortWatch	Tanker flows, port calls, chokepoint transits, capacity-weighted activity	Physical trade, supply disruption, congestion, regional flow changes	Reverse causality, noisy cargo inference, missing AIS activity
Remote sensing	Night-time lights, NO ₂ , cloud cover, site-level imagery or embeddings	Industrial activity, demand conditions, inventory observability, infrastructure signals	Indirect mechanism, weak within-site temporal variation, cloud/missing data
Multimodal learning	Modality-specific representations and fusion	Preservation of heterogeneous structure before prediction	Limited direct evidence in oil-price forecasting

数据来源 / 文献	可观测信号	经济渠道	主要限制
金融与原油市场变量	滞后价格、库存、产量/炼厂活动、波动率、GPR、利率、汇率、期货/市场指标	价格持续性、不确定性、宏观金融条件、市场预期	基准很强，难以进一步改进
航运 / AIS / PortWatch	油轮流量、港口停靠、咽喉通行、运力加权活动	实物贸易、供给扰动、拥堵、区域流动变化	反向因果、货物推断有噪声、AIS 活动缺失
遥感	夜间灯光、NO ₂ 、云量、站点影像或嵌入	工业活动、需求条件、库存可观测性、基础设施信号	机制间接、站点内部时间变化较弱、云层/缺失数据问题
多模态学习	模态特定表示与融合	在预测前保留异质数据结构	在原油价格预测中缺少直接证据

2.6.2 Research gap

2.6.2 研究空白

Existing oil-price forecasting studies have made progress in both financial modelling and machine-learning methods, but most alternative-data applications still reduce heterogeneous data to engineered tabular features. Shipping activity is usually represented as counts, flows or chokepoint indicators; satellite imagery is usually represented through indices such as night-time lights, NO₂ or cloud measures; and these variables are then concatenated with financial predictors in a flat feature table.

现有原油价格预测研究已经在金融建模和机器学习方法上取得进展，但多数替代数据应用仍然会将异质数据压缩为工程化表格特征。航运活动通常被表示为计数、流量或咽喉指标；卫星影像通常被表示为夜间灯光、NO₂ 或云量等指数；这些变量随后与金融预测变量拼接为一个扁平特征表。

What remains unclear is whether the added value comes from the alternative data themselves, or from preserving their modality-specific structure before prediction. Existing studies have not systematically compared flat concatenation of financial, shipping and remote-sensing features against representation-level modality-aware fusion for weekly Brent forecasting under the same leakage-safe out-of-sample evaluation protocol. The open question is not simply whether more data improve prediction, but whether the way heterogeneous data are represented and fused affects forecasting performance.

尚不清楚的是，增量价值究竟来自替代数据本身，还是来自在预测之前保留其模态特有结构。现有研究尚未在同一个防止数据泄漏的样本外评估框架下，系统比较金融、航运和遥感特征的扁平拼接与表示层面的模态感知融合对周度 Brent 价格预测的影响。待回答的问题并不只是“更多数据是否改善预测”，而是“异质数据如何被表示和融合，是否会影响预测表现”。

2.6.3 Positioning of this dissertation

2.6.3 本文定位

This dissertation is positioned as an empirical integration and comparison study rather than a proposal of a new neural architecture. It brings together three strands of literature: the oil-forecasting literature's emphasis on strong baselines and rigorous out-of-sample testing; the alternative-data literature's use of shipping and satellite proxies for economic activity; and the multimodal-learning literature's distinction between flat feature fusion and representation-level modality-aware fusion.

本文定位为一项经验性整合与比较研究，而不是提出一种新的神经网络结构。它结合了三类文献：原油预测文献对强基准和严格样本外检验的强调；替代数据文献将航运和卫星信号作为经济活动代理变量的做法；以及多模态学习文献对扁平特征融合与表示层模态感知融合的区别。

The dissertation therefore asks three linked questions:

因此，本文提出三个相互关联的研究问题：

- **RQ1:** Do remote-sensing and shipping indicators add incremental out-of-sample value over a financial baseline and the random-walk benchmark?
- **RQ1:** 遥感和航运指标是否能在金融基准模型和随机游走基准之上提供样本外增量预测价值？
- **RQ2:** Does modality-aware representation-level fusion outperform flat feature fusion when both use the same underlying data?
- **RQ2:** 在使用相同底层数据的情况下，模态感知的表示层融合是否优于扁平特征融合？

- **RQ3:** Can modality-level interpretability reveal which signals the model relies on across different market conditions?
- **RQ3:** 模态层面的可解释性分析是否能够揭示模型在不同市场条件下依赖哪些信号?

Both claims — that alternative data are useful and that representation-level fusion is superior — are treated as empirical questions to be tested under a consistent evaluation framework, not as assumptions.

替代数据是否有用、表示层融合是否更优——这两个主张都被视为需要在统一评估框架下检验的经验问题，而不是本文的预设前提。

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